

## Note 15.5 Triple Integrals in Rectangular Coordinates

### Triple Integrals

$$\iiint_D F(x, y, z) dV = \iiint_D F(x, y, z) dx dy dz$$

We can see the function as the **density of unit volume** and the integral as the sum of small **mass**.

### Volume of a Region in Space

#### Definition

The **volume** of a closed, bounded region  $D$  in space is

$$V = \iiint_D dV = \iiint_D dx dy dz$$

### Average Value of a Function in Space

The average value of  $F$  over  $D$  is :

$$\text{ave} = \frac{1}{D} \iiint_D F dV$$

### Properties of Integrals

The same algebraic properties as double and single integrals.